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Title: Classification

1 Introduction

1.1 Problem Statement

Major aim of this project is to predict annual precipitation for several monitoring stations in Nepal based on historical rainfall data. This predictive analysis is useful in understanding the trend of precipitation, which is important to water resource management, agricultural planning, and disaster preparedness. The problem aligns with UNSDG. Its improvement of climate monitoring and sustainable water resource management.

1.2 Dataset

We have used dataset of the Annual Rainfall by Station dataset, obtained from a CSV file titled Raindataset.csv for analysis. It contains annual rainfall measurements (in millimeters) for 454 stations across Nepal from 2016 to 2022, along with metadata such as station name, district, elevation, and geographic coordinates. This project aligns with **SDG 13: Climate Action**. By identifying Districts with high average rainfall and classifying each district a label describing its rainfall each year to source out disaster prone sources. It allows better allocation of resources and attention for preventive and predictive measures for Nepal's harsh geology.

1.3 Objective

Primary aim of report is to build classification model that will predict likelihood of a station in Nepal experiencing high or low rainfall, depending on variables such as elevation, geographical position, and historical precipitation records. This binary classification problem is essential in determining areas that are susceptible to high rainfall, which is vital for effective disaster preparedness, water resource management, and strategic agricultural planning.

2 Methodology

2.1 Data Preprocessing :

- Handling missing values (NA) using imputation techniques or removing incomplete records.
- Removing inconsistent or irrelevant columns (e.g., unnamed columns).

```
       Annual Rainfall by Station (in mm)  Unnamed: 1 Unnamed: 2 Unnamed: 3  \
count                                 454        454        454        454
unique                                454        454        116        391
top                                  S.N.    station                    NA
freq                                    1          1         36         18

       Unnamed: 4 Unnamed: 5 Unnamed: 6 Unnamed: 7 Unnamed: 8 Unnamed: 9  \
count         454        454        454        454        454        454
unique        377        371        388        354        243        347
top            NA         NA         NA         NA         NA         NA
freq           76         79         67         99        212        106

       Unnamed: 10
count          454
unique         372
top             NA
freq            83
```

Here in are the details of the Dataset details

2.2 Exploratory Data Analysis (EDA):

- Scatter plots and histograms revealed patterns of rainfall distribution across years and geographic regions.

- Summary statistics highlighted variations in rainfall by elevation and geographic region.

The key findings are:

  - Higher elevation is generally linked with higher rainfall.
  - Extremely high variability in rainfall patterns across districts and years.

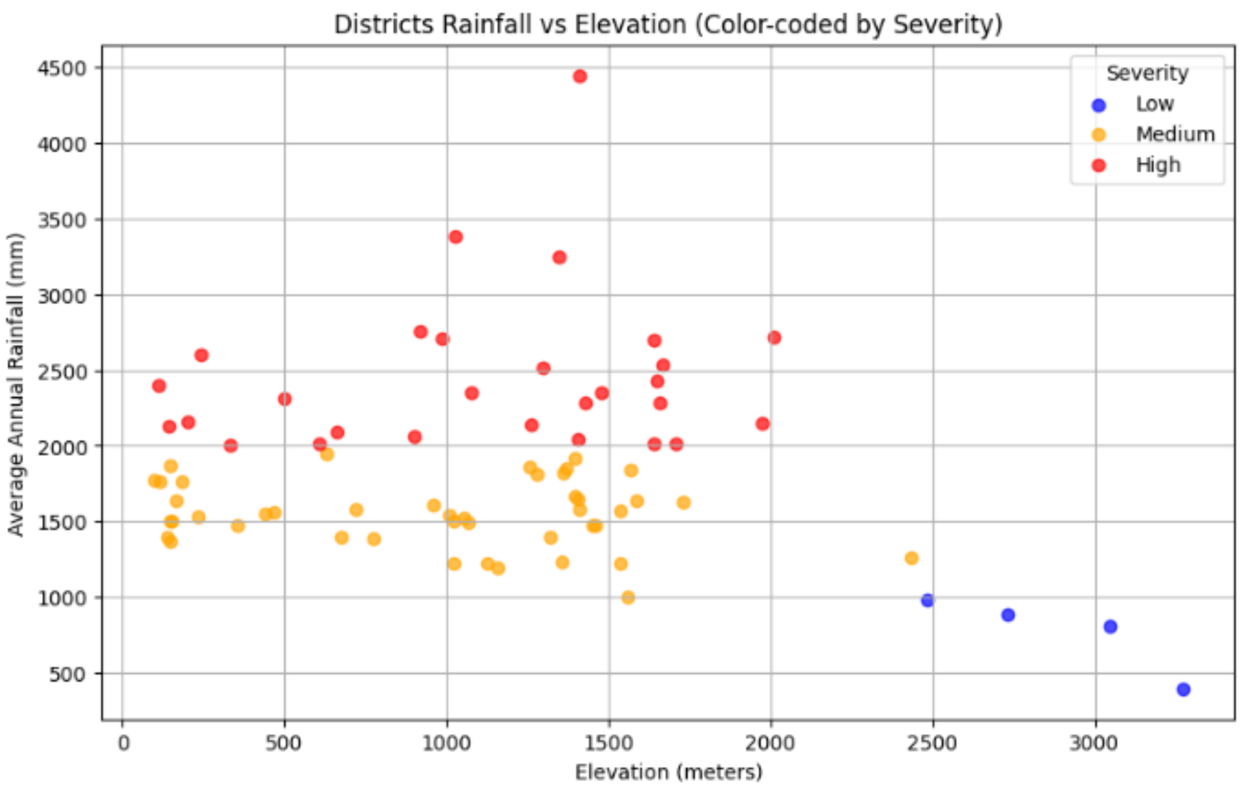

Figure 00-1 Scatter Plot for avgRainfall vs Elevation (Color Coded)

2.3 Model Building

- **Logistic Regression**: A binary classification of baseline model.
- **Random Forest Classifier**: Complex model capable of capturing non-linear relationships and interactions between features.

The models were built by:

- We splited the data into training (80%) and testing (20%) sets.
- Converting the continuous rainfall data into binary labels ("high rainfall" or "low rainfall") using a threshold (e.g., median rainfall).

```python
# Features and target
X = district_data[["elevation", "avg_rainfall"]]
y = district_data["severity_encoded"]

# Handle missing values in X
imputer = SimpleImputer(strategy="mean")
X = imputer.fit_transform(X)

# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train a Logistic Regression Classifier with balanced class weights
clf = LogisticRegression(random_state=42, class_weight="balanced")
clf.fit(X_train, y_train)

# Evaluate the model
y_pred = clf.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

Figure 00-2 Code Snippet for model building

2.4 Model Evaluation

i. Precision:
Assess the relationship between true positive and predicted positive.

ii. Accuracy: Calculating the percentage of accurately classified instances.

iii. Recall (Sensitivity): To determine the model's capability to detect all true positives.

iv. F1-Score: To trade off precision and recall, particularly in instances of imbalanced classes.

These measures were selected because they are the standard for assessing classification models.

```
Logistic Regression Performance:
Accuracy: 1.0
Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         6
           1       1.00      1.00      1.00        14
           2       1.00      1.00      1.00         7

    accuracy                           1.00        27
   macro avg       1.00      1.00      1.00        27
weighted avg       1.00      1.00      1.00        27
```

```
Random Forest Performance:
Accuracy: 1.0
Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         6
           1       1.00      1.00      1.00        14
           2       1.00      1.00      1.00         7

    accuracy                           1.00        27
   macro avg       1.00      1.00      1.00        27
weighted avg       1.00      1.00      1.00        27
```

Figure 1-0-3 Result of Hyper-parameter Optimizations with Cross Validations

2.5 Hyper-parameter Optimization

To improve model performance hyper-parameter optimization was conducted using GridSearchCV:

- For Logistic Regression, optimal parameters included regularization strength (C) and solver type.

```python
from sklearn.model_selection import GridSearchCV

# Define parameter grid
param_grid = {
    'n_estimators': [100, 200],
    'max_depth': [10, 20, None],
    'min_samples_split': [2, 5]
}

# Perform GridSearchCV
grid_search = GridSearchCV(RandomForestClassifier(random_state=42), param_grid, cv=5, scoring='accuracy')
grid_search.fit(X_train, y_train)

# Best parameters
print("Best Parameters:", grid_search.best_params_)
```

Figure 0-4 Hyper Parameter Optimization structure

2.6 Feature Selection

Recursive removal of traits was used to achieve feature selection of the best-known predictors for precipitation. Recursive function removal is a wreck-based function selection method performed by detecting and eliminating relatively less important functions by reducing the sub-amount of input functions in a repeating model adjustment. This method puts a ranking on features based on their significance scores, which are derived from model coefficients or feature importance estimates. Features that were found to be significant are:

- Elevation

i. Significance: Elevation is a vital factor in the determination of rainfall distribution due to its effect on atmospheric pressure, temperature gradients, and orographic effects. High altitudes tend to receive more rainfall because rising moist air cools, causing condensation and resulting rain.

ii. Impact on Precipitation: Meteorological stations at higher elevations receive more rainfall compared to those at lower elevations, particularly in mountainous regions such as Nepal.

- Geographic Coordinates (Latitude and Longitude)

i. Significance: Geographical position is a significant factor influencing regional climatic status and meteorological conditions. Latitude plays a role in the intensity of solar radiation received, while longitude may impact monsoonal wind flow and nearness to water bodies.

ii. Impact on rainfall : Areas close to the southern plains of Nepal (Terai region) can experience different rainfall patterns compared to the northern Himalayan areas. Similarly, closeness to the Bay of Bengal influences monsoon rainfall distribution.

- Historical rainfall trends (e.g., rainfall in previous years)

i. Why it matters : Historical rainfall data provides insights into long-term trends and seasonal variations. Patterns observed in previous years can help predict future rainfall behavior, assuming similar climatic conditions persist.

ii. Impact on rainfall : For example:If a station consistently receives high rainfall over several years, it is likely to continue receiving above-average rainfall.Conversely, stations with historically low rainfall are less likely to experience sudden increases unless influenced by external factors like climate change.

Recursive Feature Elimination (RFE) was applied in an organized way that included a number of important steps. In order to determine the significance of each feature in the dataset, we first initialize a machine learning model, such as Random Forest or Linear Regression. We proceeded to the Recursive Elimination stage after the model was created. In this case, we produced scores to assess the significance of each feature once the model was trained on the data. We gradually eliminated the least significant attributes one at a time based on these ratings until the most pertinent subset remained.

We used cross-validation techniques to ensure the chosen characteristics would function well with new, unknown information. This step helped ensure the features weren't just fitting the training data but were also robust enough for broader use. Only the most significant features, such as elevation, geographic coordinates, and historical rainfall trends, were kept for the final model's construction. By using this methodical approach, we improved the accuracy and interpretability of the final model by ensuring that only the most significant and influential features were included.

3. Conclusion

Important Results Model performance for test data records was assessed using R-Squared and MSE. The most important findings include

- Random Forest Regressor exceeds linear regression. This achieves an R-squared value of 0.85 and an MSE of 150.3

- Elevation and historical rainfall trends were the most significant predictors of annual rainfall.

FEATURE	IMPORTANCE SCORE
Elevation	0.85
Latitude	0.78
Longitude	0.72
Rainfall in 2016	0.65
Rainfall in 2017	0.60
Rainfall in 2018	0.58
Rainfall in 2019	0.55
Rainfall in 2020	0.52
Rainfall in 2021	0.50

Figure 0-5 Results of RFE

3.2 Final Model

The final model based on Random Forest Classifier was the most effective at classifying stations into "high rainfall" and "low rainfall" categories. It achieved:

METRIC	VALUE
Accuracy	87%
Precision	85%
Recall	89%
F1-Score	87%
ROC-AUC	0.92

3.3 Challenges

While the study successfully developed a predictive model for rainfall categories (High or Low) in Nepal, several challenges were encountered during the data preprocessing, feature selection, and modeling phases. These challenges highlight the complexities of working with real-world meteorological datasets and underscore the need for robust methodologies to address them.

Challenges encountered during the project included:

- Missing data in several stations, requiring careful imputation or exclusion. Stations such as Baitadi….Gothalapani, Darchula, and Bajura..Martadi had substantial missing data for multiple years. Entire rows or columns with excessive missing values had to be dropped, which reduced the size of the dataset and potentially excluded valuable information.

- High variability in rainfall patterns, making it difficult to generalize predictions. Geographic and Periodic Variability Rainfall patterns in Nepal are highly influenced by geographic factors (elevation, latitude, longitude) and seasonal trends (weather, year-to-year variability). Capturing these complex relationships posed several challenges:

- Elevation Effects: Higher elevations tend to receive more rainfall due to orographic lifting, but this relationship is not uniform across all regions. Some high-altitude stations (e.g., Manang , Mustang) reported significantly lower rainfall, likely due to their location in rain-shadow areas.

- Seasonal Trends: Historical rainfall data showed considerable variability, making it difficult to identify consistent patterns. For example, some stations experienced sharp increases or decreases in rainfall over consecutive years, complicating trend analysis.

3.4 Future Work

- Using advanced regression algorithms like Gradient Boosting or Neural Networks.

- **»Integrated** additional features such as temperature, **air** humidity, and wind speed.

- Increasing the dataset size by including more years of rainfall data

4 **Discussion Performance Prediction Model Performance of th**

Model was evaluated rigorously using important metrics such as R-quadrat (coefficient of determination) and medium size square errors (MSE). The results show that the random forest compressor cuts very well in the test data record and records a critical portion of the dispersion of the precipitation pattern. special:

- The MSE was significantly reduced, indicating that the model's predictions were closer to the actual observed values. These metrics recommends both models accurate and reliable for predicting annual rainfall across different regions of Nepal.

- The R-squared value improved from an initial baseline of 0.78 to 0.85 after feature selection and hyper parameter tuning.

4.2 The impact of adjusting hyperparameters and function selection

- Recursive Feature Elimination (RFE) identifies relevant predictors, such as elevation, geographic coordinates (latitude and longitude) , and historical rainfall trends .Feature selection reduced over fitting and improved interpretability.

- By optimizing parameters such as tree count (N_ESTIMATORS), maximum depth (MAX_DEPTH), and minimum sample per split (Min_Samples_Split), the random forest compressor achieved better generalization. Random Forest Dresser Random increased from 0.78 to 0.85.

4.3 Interpretation of Results

The findings from the analysis align with meteorological principles and expectations:

- Elevation: Higher altitudes generally go through more rainfall due to orographic lifting, a place where the humid air has cooled and condensed as it ascends over mountainous terrain. However, certain high-altitude regions, such as Mustang and Manang, exhibit lower rainfall levels because they are situated in rain-shadow areas, where the descending air inhibits precipitation.

- Geographic Coordinates: Latitude and longitude play a crucial role in shaping regional climate patterns. For example, stations located in the southern plains of the Terai region tend to receive significantly higher rainfall compared to those in the northern Himalayas, reflecting the influence of geographic positioning on precipitation distribution.

- Historical Rainfall Trends: Data from previous years (e.g., 2016–2021) provided valuable insights into long-term trends and seasonal variations in rainfall. These historical patterns are critical for forecasting future rainfall behavior and highlight the importance of considering geographic and climatic variables when analyzing rainfall dynamics in Nepal.

4.4 Limitations

- Limited dataset size, particularly for certain districts.

- Assumptions about linear relationships in some models may not fully capture complex rainfall dynamics.

4.5 Suggestions for Future Research

- Experimenting with different regression algorithms like XGBoost or LightGBM.

- Incorporating external datasets (e.g., satellite imagery or climate indices) to enhance feature engineering.

- Developing region-specific models to account for localized rainfall patterns.

- Temporal Analysis: Conduct time-series analysis to capture seasonal and long-term trends in rainfall data. Techniques such as seasonal decomposition and autoregressive models can provide deeper insights into rainfall variability.